

BAFI 6003 - Predictive and Visual Analytics

2026 S1, Individual Assignment

March 25, 2026

Instructions

1. This is an individual assignment.
2. The maximum score is 25 points.
3. The presentation of your write-up is important. Poorly formatted reports might lose up to 5 points.
4. All numerical analysis, all tables and figures need to be done using Tableau or Python (however, in Task 2, you may use Excel or Word etc. to make tables for regressions).
5. Please retain your Python code and make sure that it is user-friendly (use comments where necessary). Using your submitted code, one should be able to produce all your results, tables, and figures.
6. Please retain a copy of the problem set that is submitted.
7. You should submit the following files (please do not 'zip' them):
 - the Tableau dashboard for Task 1
 - the report (in doc, docx, or pdf format) for Tasks 2 and 3; the report should be properly formatted and be similar to a business report; font: 12 pt Times New Roman; maximum number of pages: 10 (no penalty for exceeding this limit)
 - Python code (in py format or Jupyter notebook, or Google Colab based .ipynb document) that can be used to replicate all the results in Tasks 2 and 3.

8. Lecturer can refuse to accept assignments, which do not have a signed acknowledgment of the University's policy on plagiarism.
9. Any suspected plagiarism will be severely punished. This includes any student that submits copied work or any student that allows their work to be copied.
10. You must acknowledge any external material you use in your answers, e.g., material from websites, textbooks, academic journals, newspaper articles, or generative AI tools.
11. All queries (including deadline extensions) for this project should be directed to Course Coordinator.
12. The submission deadline for the Individual Assignment is 6pm, Friday the 17th of April, 2026.
13. The submission must be done through Canvas site.
14. Students who submit their assignment after the published deadline, without an approved extension, will have 1.25 points deducted for every day (or part thereof) the assignment is overdue, for a maximum of 7 calendar days. After 7 calendar days, the assignment will not be accepted for marking.

Academic Honesty Statement

By submitting an assignment via Canvas students are agreeing to the following statement:

'I declare that all material in this assessment is my own work, except where there is clear acknowledgement and reference to the work of others. I have read the Adelaide University's Academic Honesty Policy. I give permission for any assessed assignments to be reproduced and submitted to other academic staff for the purposes of assessment and to be copied, submitted and retained in a form suitable for electronic checking of plagiarism.'

Agenda

Assume that you are an equity analyst working at a reputable investment bank. Your client is a large multinational corporation and would like to hire your bank for

a seasoned equity issue (SEO).¹ Your goal is to set the fee for this issue. A fee is in per cent of the issue size. For example, if SEO size is \$100 million and fee is 2%, then the investment bank will get \$2 million from the issuer.

You have been provided with a data set (file name: SDC) of all SEOs conducted in the US. It contains the following variables:

- CUSIP – issuer code
- MASTER_DEAL_NO – issue code
- NINAMES – issuer name
- STATE – issuer’s state
- PRINCIPALAMOUNT – issue size in \$ millions
- EXCHANGE – stock exchange where the stock of issuer is traded
- INDUSTRY – industry description
- MKTVALUEPF – market capitalization of the issuer after an SEO in \$ millions
- INDUSTRY_CODE – industry code
- FEES – SEO fees in %
- OFFER_PRICE – the price of newly issued stock per share.

First, you should prepare your data for the analysis:

- remove duplicates (if any)
- remove observations with missing values of FEES variable
- check for outliers and take necessary actions to deal with them
- and so on.

¹An SEO is an equity issue of a firm which stock is already listed on a public stock exchange. In other words, if a public firm would like to sell additional shares to investors, it conducts an SEO (let’s say a firm needs to raise money in order to build a new plant). To do this, a firm hires an investment bank (which is called underwriter) that helps to promote the issue, find large institutional investors which could buy its large portion, set the initial price for newly issues shares (in most cases, they would be issued at a small discount compared to the actual stock price) and so on.

1 Dashboard (10 points)

This task needs to be done using Tableau.

Create a dashboard focusing on SEO issue costs (FEES). While the design and layout are entirely at your discretion, the dashboard must meet the following requirements:

- The dashboard must include four charts.
- One chart must be a geographic map of the US.
- The dashboard must feature a year-based animation.
- The dashboard should feature a filter.

2 OLS regressions (12 points)

This task needs to be done using Python.

Discuss briefly your sample, including the number of observations, outliers. Provide the descriptive statistics of the sample. How you choose to do this is entirely at your discretion. However, it is recommended that you consider using both summary statistic and graphical methods (this task should include at least one properly formatted table, one pie chart, one histogram, and one scatter plot) while also noting any peculiarities within the data set. You should put more emphasis on FEES and PRINCIPALAMOUNT. Tables and figures should be included in the report rather than appendix.

Find the determinants of the SEO issue fees (FEES). Estimate 3 different OLS regressions (in order to ensure the robustness of results) with year and industry (use INDUSTRY_CODE variable) fixed effects and several independent variables. To ensure that regression residuals ‘behave well’, you may need to scale or transform one or more variables. For example, to use a natural logarithm value of the variable instead of its raw value. Provide a properly formatted table with the regression results in the report (not in the appendix; however, you may put additional tables in the appendix if needed). Discuss the determinants of the SEO issue costs: what variables are statistically significant; which variables increase and which variables decrease issue costs; any insights from the coefficient estimates of year and industry fixed effects and so on.

Predict the SEO issue fees using the results from the 3 regressions for the following issue:

- Year=2025
- issue size= \$100 million
- EXCHANGE = 'New York'
- market capitalization of the issuer after an SEO = \$5 billion
- OFFER_PRICE = \$50
- INDUSTRY_CODE = 3.

Are the predictions similar across the 3 models?

3 Time series analysis (3 points)

This task needs to be done using Python (and partially a generative AI tool).

Generate a time series values for SEO issue costs (FEES); that is, annual averages for each year. Plot the obtained time series. Use ARMA type models to predict its values for the next 2 periods. Motivate and discuss ARMA orders used in the analysis. Plot the fitted (predicted) values:

- a scatterplot against the actual values (for the 1995-2025 period)
- time series plots of the actual and fitted (predicted) values in the same figure (for the 1995-2027 period).

Do actual and fitted (predicted) time series tend to move in the same direction over time?

Your client considers conducting the following SEO this year (i.e., in 2026):

- Year=2026
- issue size= \$100 million
- EXCHANGE = 'New York'
- market capitalization of the issuer after an SEO = \$5 billion

- $\text{OFFER_PRICE} = \$50$
- $\text{INDUSTRY_CODE} = 3$.

What fee would you quote them, and why? To answer this question, utilize a generative AI tool (e.g., ChatGPT, Google Gemini). Prompt your chosen AI using the results from this and previous tasks, and instruct it to generate an answer of no more than 200 words.

In your report, you must include the exact response provided by the AI and critically discuss whether you agree with its assessment. If you do not fully agree with the AI's solution, provide and justify your own fee quotation.

Good luck!